

Economic Predictions with Big Data: The Illusion of Sparsity

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- Recent availability of large datasets have generated interest in predictive models with many possible predictors.
- OLS, MLE, or Bayesian inference perform poorly, since the proliferation of regressors magnifies estimation uncertainty and produces inaccurate predictions.
- Inference methods aimed at dealing with the curse of dimensionality can be divided into two broad classes:
 - Sparse-modeling
 - Dense-modeling

- Are economic predictive problems more likely characterized by sparsity or density?
- Address this question using a linear predictive model and a "spike-and-slab" prior
- Estimate this model on six large datasets from macroeconomics, finance, and microeconomics
 - These applications exhibit substantial variation, spanning time-series, cross-section, and panel data, different number of predictors, and predictors-to-observations ratios
- Model uncertainty is pervasive, and ignoring it leads to an "illusion of sparsity"

Three Key Ingredients of the Model

- 1 It allows for the possibility of sparsity by assuming that some regression coefficients may be equal to zero.
- 2 It shrinks the non-zero coefficients towards zero, as an alternative way to reduce estimation uncertainty and avoid overfitting for high-dimensional models.
- 3 It treats the degree of sparsity and shrinkage separately, as each are controlled by different hyperparameters, q and γ^2

Consider a linear model to predict a response variable y_t :

$$y_t = u_t' \phi + x_t' \beta + \epsilon_t$$

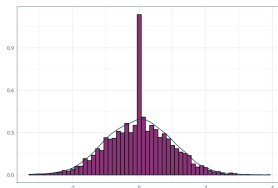
$$\epsilon_t \sim i.i.d. \quad N(0, \sigma^2)$$

u_t represents the set of explanatory variables that a researcher always wants to include in the model

x_t represents possibly but not necessarily useful predictors of y_t

Hyperparameters

- q is the probability that regression coefficients can be non-zero and are included in the model, and determines the degree of sparsity
 - q higher $\implies$ denser
 - q lower $\implies$ sparser
- γ^2 controls the degree of shrinkage given a predictor is included. The higher γ^2 , the less shrinkage is performed
- "Spike-and-slab" prior



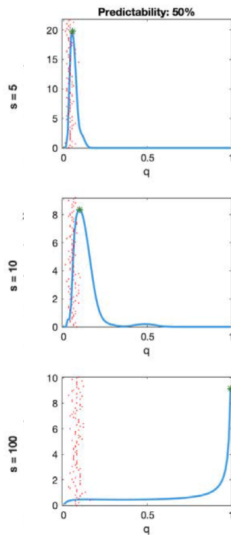
Prior distribution for the unknown coefficients:

$$p(\sigma^2) \propto \frac{1}{\sigma^2}$$

$$\phi \sim flat$$

$$\beta_i | \sigma^2, \gamma^2, q \sim \begin{cases} N(0, \sigma^2 \gamma^2) & \text{with pr. } q \\ 0 & \text{with pr. } 1 - q \end{cases} \quad i = 1, \dots, k$$

Testing the Model



	Dependent variable	Possible predictors	Sample
Macro 1	Monthly growth rate of US industrial production	130 lagged macroeconomic indicators	659 monthly time-series observations, from February 1960 to December 2014
Macro 2	Average growth rate of GDP over the sample 1960-1985	60 socio-economic, institutional and geographical characteristics, measured at pre-1960s value	90 cross-sectional country observations
Finance 1	US equity premium (S&P 500)	16 lagged financial and macroeconomic indicators	68 annual time-series observations, from 1948 to 2015
Finance 2	Stock returns of US firms	144 dummies classifying stock as very low, low, high or very high in terms of 36 lagged characteristics	1400k panel observations for an average of 2250 stocks over a span of 624 months, from January 1963 to May 2014
Micro 1	Per-capita crime (murder) rates	Effective abortion rate and 284 controls including possible covariate of crime and their transformations	576 panel observations for 48 US states over a span of 144 months, from 1986 to 1997
Micro 2	Number of pro-plaintiff eminent domain decisions in a specific circuit and in a specific year	Characteristics of judicial panels capturing aspects related to gender, race, religion, political affiliation, education and professional history of the judges, together with some interactions among the latter, for a total of 138 regressors	312 panel observations for 12 circuits over a span of 26 years, from 1979 to 2004

TABLE 1. Description of the datasets.

Positive Correlation b/w Inclusion Probability & Shrinkage Degree

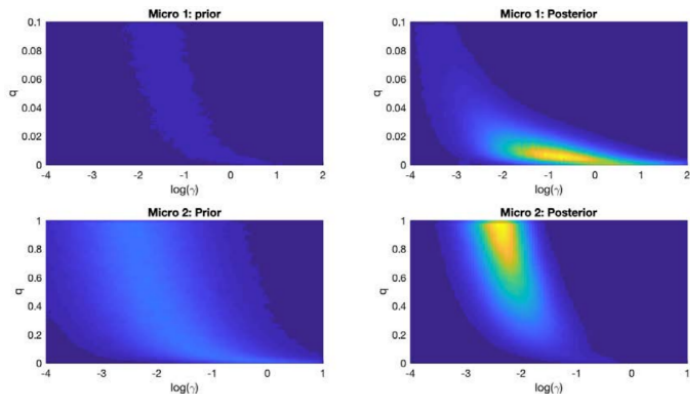
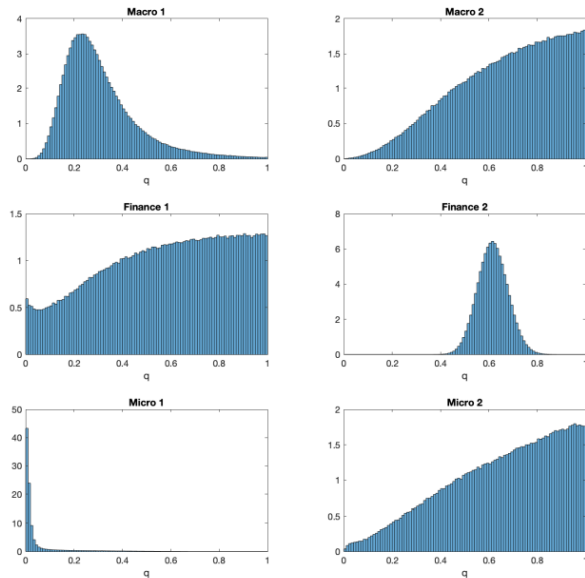


FIGURE 4.2. Joint prior and posterior densities of q and $\log(\gamma)$ in the finance-2, micro-1 and micro-2 applications (best viewed in color).

Probability of Inclusion and Degree of Sparsity



Probabilities of Inclusion of Each Predictor

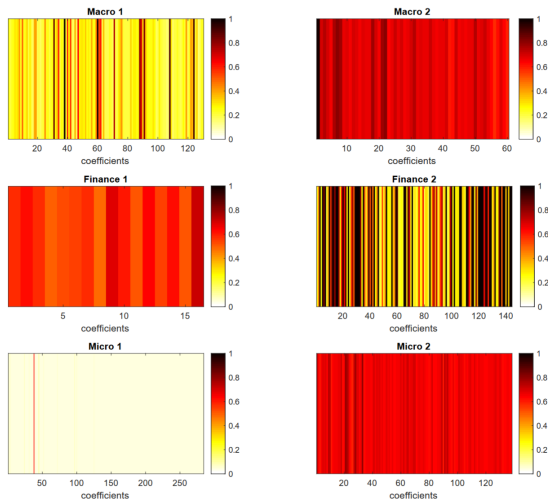


FIGURE 4.4. Heat map of the probabilities of inclusion of each predictor (best viewed in color).

- There is no theoretical argument suggesting that predictive models should in general include only a handful of predictors.
- The empirical support for low-dimensional models is generally weak.
- Predictive uncertainty is too pervasive to ignore.
- The right approach is to assess and fully convey this uncertainty, rather than understating it through the use of dogmatic (prior) assumptions favoring low dimensional models.